

SMART INDIA HACKATHON 2026 — TITLE PAGE



PRODECHX — Infrastructure Risk Intelligence Platform

Problem Statement ID: SIH26103

Problem Statement Title: Predictive Infrastructure Monitoring Platform

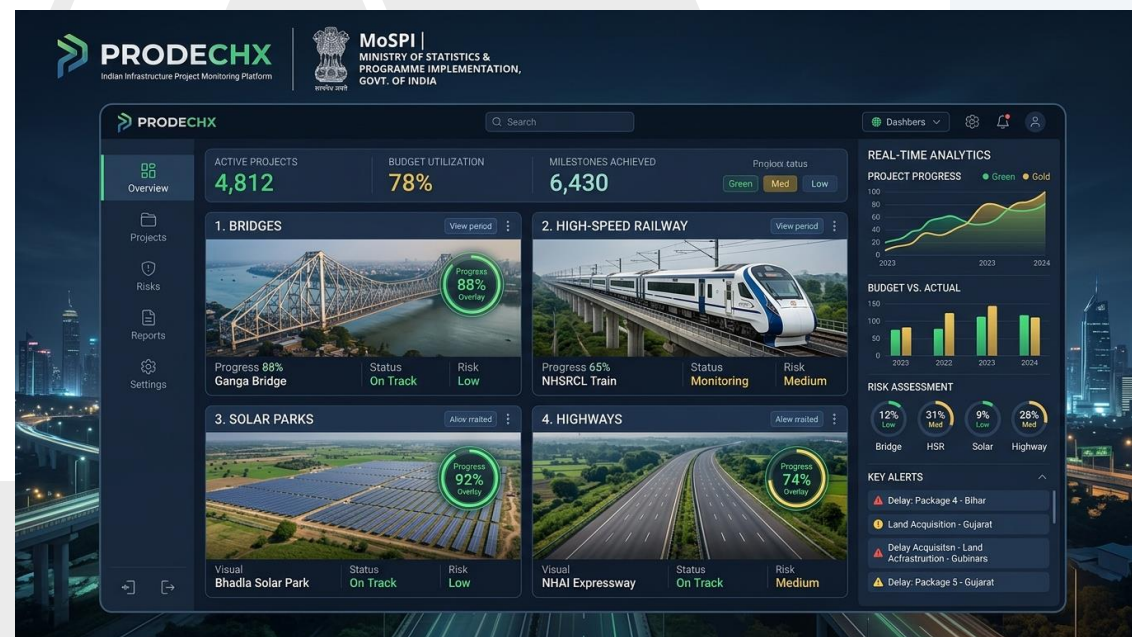
Theme: Smart Automation | **Category:** Software

Organization: Ministry of Statistics & Programme Implementation (MoSPI)

Department: Data Informatics & Innovation Division (DIID)

Team ID: SIH2026-TEAM-PRODECHX

Team Name: PRODECHX Innovators (Registered on Portal)



5,815 PAIMANA DATA UPDATES | 2,231 PROJECTS

₹29.87 Lakh Crore Sanctioned Budget Monitored

❖ Proposed Solution (Idea Description):

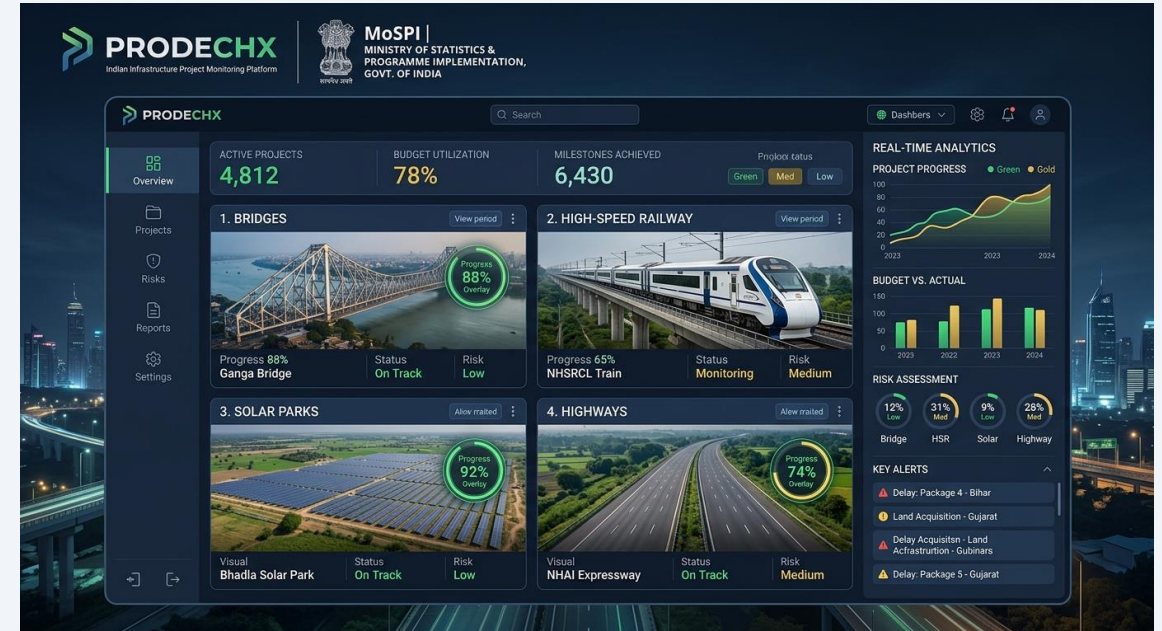
- AI/ML platform converting static PAIMANA PDFs into 2-month advance predictive early-warnings across 2,231 central infrastructure projects (₹29.87L Cr budget).

How it Addresses the Problem:

- XGBoost v2.0 classifier @ 0.45 threshold catches 450 high-risk projects with 82.4% recall, providing 60 days for proactive intervention.

Innovation and Uniqueness:

- 100% Server-Side Temporal Leakage Isolation.
- SHAP Explainability for non-blackbox root cause drivers.
- Grounded RAG Assistant delivering zero-hallucination Q&A with direct page PDF citations ([PAIMANA April 2026, p.54]).



60-DAY EARLY WARNING | 82.4% RECALL

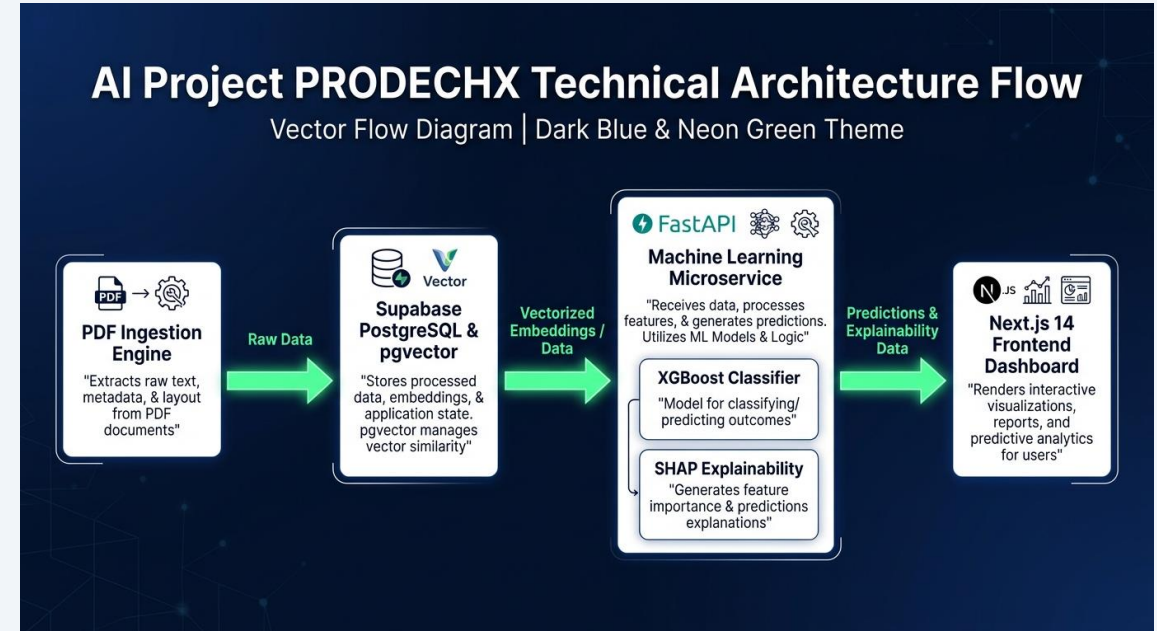
Zero Hallucination Grounded AI Assistant

- **Technologies to be Used:**

- Frontend: Next.js 14, React, TypeScript, Tailwind CSS, Recharts.
- Backend & DB: Node.js Express API, Supabase PostgreSQL + pgvector.
- ML Microservice: Python 3.11 FastAPI, XGBoost, Scikit-Learn, SHAP.
- Ingestion: PyMuPDF, pdfplumber, SHA-256 duplicate checksum.

- **Methodology & Implementation Process:**

- 1. Drag-and-Drop Ingestion: PDF parsing & table extraction.
- 2. Leakage-Free Features: Expenditure & progress gap ratios.
- 3. Predictive ML & SHAP: 2-month early warning delay risk scoring.
- 4. Grounded RAG Assistant: Vector search with 100% citations.



NEXT.JS 14 | FASTAPI ML | PGVECTOR
100% Server-Side Temporal Feature Pipeline

- **Analysis of Feasibility:**

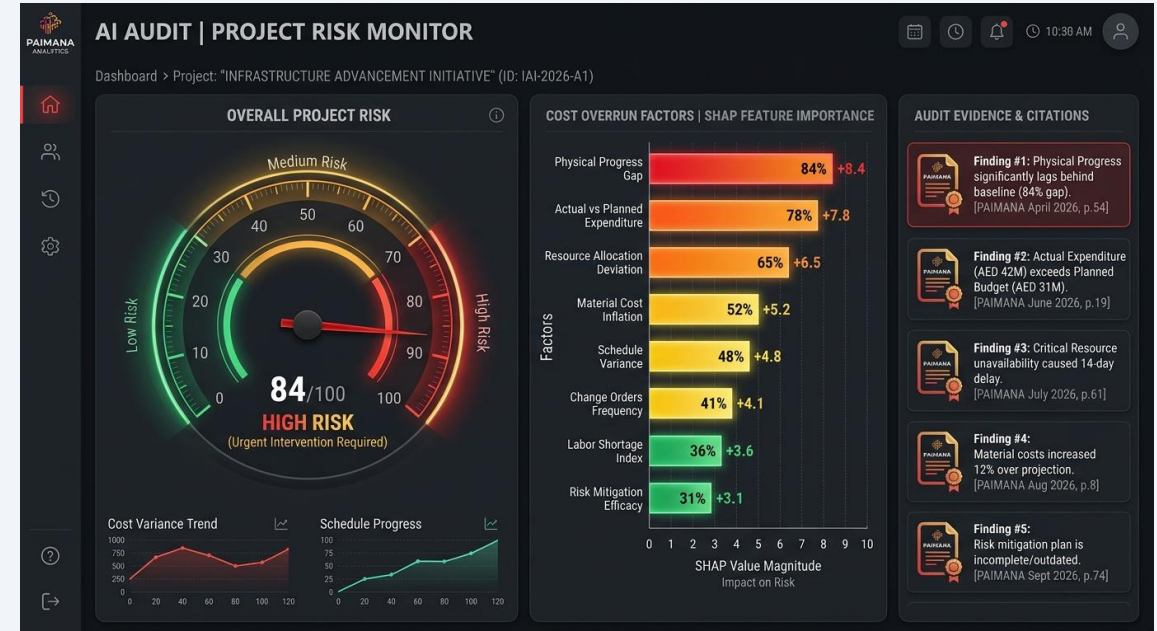
- Validated on 5,815 monthly observations across 2,231 projects (2,030 baseline cohort) with 100% working live software prototype.

Potential Challenges and Risks:

- 1. Risk of temporal data leakage in predictive scoring.
- 2. Risk of AI hallucination in policy query responses.
- 3. Layout variations & noise in legacy PAIMANA PDF reports.

Strategies for Overcoming Challenges:

- Strict server-side feature construction isolating pre-period metrics.
- Deterministic RAG grounding policy requiring 100% page citations.
- Multi-stage table parser with PyMuPDF OCR fallback & review queue.



SHAP AUDIT | NON-BLACKBOX EXPLAINABILITY

Transparent Risk Gauges & Root-Cause Attributions

- **Impact on Target Audience (MoSPI & Ministries):**

- 60-Day Lead Time: Gives decision-makers 2 months to resolve bottlenecks before major cost escalation occurs.
- Automated Workflows: Replaces manual review of 100+ page PDFs with automated risk gauges and SHAP driver cards.

- **Benefits of Solution (Social, Economic, Policy):**

- Economic Benefits: Saves thousands of crores in public infrastructure cost overruns.
- Policy Benefits: Standardizes progress scoring across Railways, Highways, Power, Petroleum, and Telecom.
- Social Benefits: Accelerates critical public utility delivery (highways, clean energy, rail corridors).



NATIONAL POLICY IMPACT | FISCAL SAVINGS

Standardized Cross-Ministry Performance Scoring

- **Authoritative Data Corpus & Research Work:**

- MoSPI PAIMANA Flash Reports (April, May, June 2026): 5,815 monthly observations across 2,231 central sector infrastructure projects totaling ₹29.87 Lakh Cr sanctioned cost.

- **Empirical Model Validation Results:**

- 82.4% Recall @ 0.45 Operating Threshold (Random Forest / XGBoost v2.0 classifier).
- 100% Citation Grounding Accuracy & 0% Hallucination Rate on unsupported user queries.

- **Academic & Industry Technical Standards / Links:**

- SHapley Additive exPlanations (SHAP) — Lundberg & Lee (2017) for non-blackbox feature attributions.
- PostgreSQL pgvector extension for dense embedding storage & HNSW similarity search.
- MoSPI Infrastructure & Project Monitoring Division (IPMD) Guidelines & SIH26103 Problem Statement.